

International Conference on Machine Learning and Data Engineering

Machine Learning Techniques for Breast Cancer Prediction

Varsha Nemade^{1,*}, Vishal Fegade²^aDepartment of Computer Engineering, Mukesh Patel School of Technology Management and Engineering, NMIMS Shirpur Campus, India^bDepartment of Mechanical Engineering, Mukesh Patel School of Technology Management and Engineering, NMIMS Shirpur Campus, India

Abstract

Breast cancer is main reason for mortality in woman. Prediction of breast cancer is a challenging task in medical data analysis. Doctors and pathologist required some automated tools to take decision and to differentiate between malignant and benign tumour. A machine learning (ML) algorithm helps lot to take decisions and to perform diagnosis from the data collected by medical field. Various researches show that ML techniques are helpful for decision making in breast cancer prediction. In this paper, we used various ML Classification techniques: Naïve Bayes(NB), Logistic regression (LR), Support vector machine(SVM), K-Nearest Neighbor (KNN), Decision Tree(DT), and ensemble techniques: Random forest(RF), Adaboost, XGBoost on breast cancer dataset and evaluated by using different performance measure. It has been found that both decision tree and XGBoost classifier has highest accuracy 97% among all model and highest AUC 0.999 obtained for XGBoost classifier.

© 2023 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the scientific committee of the International Conference on Machine Learning and Data Engineering

Keywords: Breast Cancer; Ensemble Techniques ;Machine Learning;

1. Introduction

GLOBOCAN 2020 report shows that how cancer is more dangerous as it recorded total 19.3 cases in that 10 million deaths in 2020 [1,2]. Female breast cancer has recorded with 2.3 million new cases [1] and also indicates the mortality rate also increased. Breast cancer created huge impact on the life of women. Mortality rate can be decreased by increasing awareness, early prediction and diagnosis [3]. Today medical field generates large amount data of different diseases and that data helps to perform analysis and further predictions. Technology helps lot to

* Corresponding author. +91942361554.

E-mail address: vvfegade@gmail.com

doctors [4] pathologist for performing an accurate prediction that helps to avoid further medical expenditure and to get proper treatment and in some cases early detection can save a person's life [5]. Different techniques have been used to classify breast cancer. Classification techniques play an important role in breast cancer prediction and diagnosis. Many research have demonstrated the importance of breast cancer prediction with different techniques and challenges [6,7], and they used a variety of data mining classification approaches to analyze the Wisconsin Diagnostic Breast Cancer (WDBC)[8,9] and Wisconsin Breast Cancer(WBC) [10]dataset and it found substantial result. Many data mining techniques [11] are used for classification such as KNN, NB, DT, SVM, LR, ensemble techniques such as RF, Adaboost, XGBoost. Our aim is to predict breast cancer by applying different classification algorithm and compare their performance and find out best algorithm. This paper organizes in following manner: Section 2 describes similar kind of work performed before on breast cancer datasets, section 3 describe the architectural overview of system with dataset, and classification techniques details, section 4 describes results and discussion of ML techniques with different performance measures.

2. Related Work

Prediction of breast cancer using ML techniques is on-going work. Research has shown that many machine learning techniques such as KNN, NB, DT, SVM, and LR have been developed for breast cancer diagnosis. Kabiraj et al. proposed a Random Forest and XGBoost system for predicting breast cancer risk and got 74.73%, 73.63% respectively[12]. Liu et al. has developed gradient boosting algorithm called EXSA to predict breast cancer progression that helps to increase survival rate [13]. Nanglia et al. has described the ensemble model using KNN, SVM, and DT with 78% [14]. Islam et al. has applied different supervised learning techniques on Wisconsin dataset and ANN got 98.57% highest accuracy [15]. Amrane et al. proposed breast cancer classification using KNN and NB using cross validation with accuracy 97.51% and 96.17% respectively [16]. Dhahri et al. proposed method that used genetic programming techniques for selection of best features and parameter for ML classifier [17]. Obaid et al. presented that SVM with quadratic kernel function achieved highest accuracy [9]. Both linear and nonlinear approaches to data reduction are viable options, and which one is used will depend on the characteristics of the correlations that exist between the features of the dataset [18]. The vast majority of ML algorithms, on the other hand, will calculate the Euclidean distance between any two data points they are given. It is necessary to bring all of the characteristics down to the same magnitude level. Scaling is a method that can be used to accomplish this goal [19]. Ara et al. performed feature selection and apply ML on breast cancer data set for classification into benign and malignant and by using SVM and RF they got 96.5% accuracy [20]. Jabbar et al. classifying the breast cancer data, an Ensemble model has been constructed in this work by utilising Bayesian networks and Radial Basis Function producing accurate classifications optimal accuracy of 97.42 % by combining these two different types of classifiers.[21]. Mahesh et al. used different machine learning techniques and also blended ensemble learning for breast cancer prediction and achieved 98.14% accuracy [22].

3. Methodology

Fig.1. shows overall design of proposed methodology applied for detection of breast cancer. Different classification algorithms applied on breast cancer data but different classifier shows different performance on same data therefore we used an ensemble technique that uses bagging and boosting which combines results from different classifier also learns from previous classifiers. To perform this ,first step of this is data acquisition. The data then pre-processed for selection of attributes, after that data divided: 80% for training and 20 % for testing. Dataset is labelled dataset having labels malignant and benign and therefore supervised different classification techniques applied on training data for building a model. Test data evaluated by using different classifier and finally compare the performance of different classifiers.

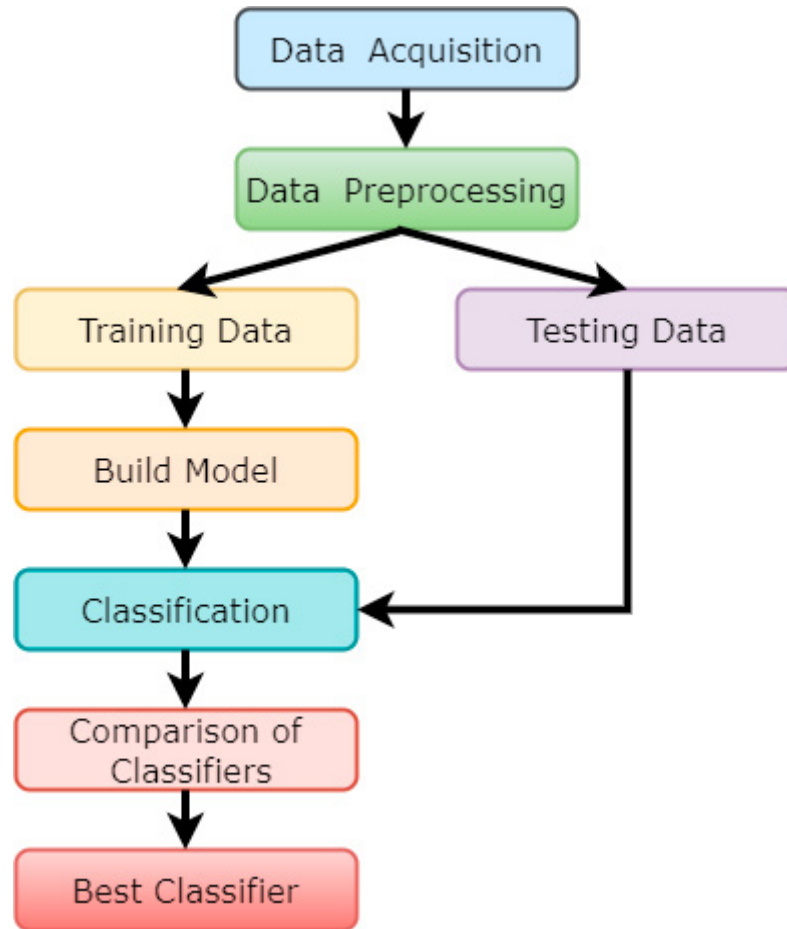


Fig.1. Proposed Flow Diagram

3.1. Dataset

For our experiments, we used the WDBC dataset. It is available on UCI repository [23]. This dataset is used to determine whether a cancer is benign or malignant. A digitized image of an affine needle aspirate (FNA) of a mass in breast is used to extract features. It describes the properties of the nuclei in the image. This dataset contains 569 total records, 212 of which are malignant and 357 of which are benign. It has characteristics such as an ID number, a diagnosis, a radius, texture, a perimeter, an area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension. There are a total of 13 features left. The mean, standard error, and worst or largest of these features are calculated for every image.

3.2. Model Building

The efficacy of medical treatment and the accuracy of the diagnosis are two factors that have a significant impact on

a patient's chances of surviving cancer and avoiding a second bout with the disease. Data is randomly selected into training and testing and split in the ratio of 80:20 respectively. The effectiveness of the model was evaluated with the help of test data after it was trained with the help of training sets. The various features values will determine whether the person will be affected or not. The first thing that needs to be done is to collect the necessary data for pre-processing in order to enhance the data's quality. This can be done by using pre-processing Data is pre-processing performed by selecting only required attributes. Separate Diagnosis column from remaining attributes, and label encoding is applied to categorical labels to convert them into numerical form. Standard scalar is applied that scales each feature to convert unit variance. After data preparation different machine models are used to train data. Once model trained its performance is evaluated on test data. In this different ML algorithms are used along with the ensemble classifier. The goal of ensemble learning is to produce a single superior predictive model by combining the results of multiple learning algorithms. It does this by combining several different kinds of supervised learners in order to boost the predictive ability of the model. Here we use Random forest bagging technique and Adaboost and XGBoost boosting techniques.

Following different Classification Algorithms and three ensemble learning techniques are applied on pre-processed breast cancer dataset to decide whether it is malignant or benign.

- Logistic Regression: It is an extension of linear regression model and used for classification problem. It provides the probabilities for two possible outcomes. In health care field this method is useful for prediction of likelihood of disease or illness.
- KNN: It uses all training data to classify new data point based on the similarity. To assign label to new data point it calculate its distance from different label point and finally it find nearest neighbour.
- SVM: This classification technique not required any prior distribution knowledge for classification. It uses hyper plane to classify data points
- Naïve Bayes: This techniques based on Bayes theorem. It performs prediction based on probability.
- Decision Tree: In this technique instances are classified using features value. For splitting it uses Gini Index, Information Gain. Leaf node indicates the label.
- Random Forest: It is ensemble technique. It creates multiple DT on different sample data generates majority votes and perform classification
- AdaBoost: Adaptive boosting is simple technique that uses decision tree. Multiple models are created; errors found in previous models are corrected in next model. It assigns weights to incorrectly classified instances and subsequent model predicts that values correctly
- XGBoost: It stands for eXtreme Gradient Boosting and based on DT algorithm and reduce overfitting.

3.3 Implementation

All machine learning algorithm used in this paper are implemented on Google Colab that provide Jupyter Notebook environment using Scikit learn library available in python. Numpy, Pandas, Matplotlib libraries are also used.

4. Results and Discussions

On the Wisconsin Breast Cancer dataset, various models are used, and their performance is measured using Accuracy and AUC, precision, and recall. We compared the performance of various models by using AUC. We divided our results into two sections: standard ML algorithms and ensemble techniques. Confusion matrix includes actual and predicted labels as well as True Negative (TN), False Negative (FN) True Positive (TP) and False Positive (FP). Precision is defined as the number of positive class predictions that are actually positive class predictions, as shown in equation 1. The recall is the number of correct positive class predictions made out of all correct positive examples in the dataset and it is calculated as shown in equation 2. The F1 Score is derived from the weighted average of Precision and Recall; it is calculated as shown in equation 3. Accuracy is calculated by using confusion matrix as shown in equation 4; it tells how the tuple in training and testing data are correctly classified.

$$Precision = \frac{TP}{TP+FP} \quad (1)$$

$$Recall = \frac{TP}{TP+FN} \quad (2)$$

$$F1\ score = 2 * (Recall * Precision) / (Recall + Precision) \quad (3)$$

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (4)$$

Table 1. Classifier Performance

Classification Technique	Parameter	Confusion matrix	Class	Precision	Recall	F1-Score	Accuracy
K-Nearest Neighbor	n_neighbors=5	[67 0] [5 42]	benign malignant	0.93 1.00	1.00 0.89	0.96 0.94	0.96
Support Vector Machine	C=5.0 Kernel=linear	[63 4] [2 45]	benign malignant	0.97 0.92	0.94 0.96	0.95 0.94	0.95
Decision Tree	criterion = "gini"	[66 1] [2 45]	benign malignant	0.97 0.98	0.99 0.96	0.98 0.97	0.97
NaiveBayes	default GaussianNB()	[61 6] [5 42]	benign malignant	0.92 0.88	0.91 0.89	0.92 0.88	0.90
Logistic Regression	solver='liblinear'	[65 2] [2 45]	benign malignant	0.97 0.96	0.97 0.96	0.97 0.96	0.96

Table 2: Ensemble Techniques Results

Classification Technique	Parameter	Confusion matrix	Class	Precision	Recall	F1-Score	Accuracy
Random Forest	n_estimators=30	[65 2] [2 45]	benign malignant	0.97 0.96	0.97 0.96	0.97 0.96	0.96
Adaboost	n_estimators=40	[64 3] [1 46]	benign malignant	0.98 0.94	0.96 0.98	0.97 0.96	0.96
XGBoost	objective="binary:logistic"	[66 1] [2 45]	benign malignant	0.97 0.98	0.99 0.96	0.98 0.97	0.97

Table 1 shows the performance of different classifier on test dataset. It shows the different classifier with their respective hyper parameters. It includes precision, recall, F1 score for each class and accuracy of the model. It indicates that decision tree classifier with highest accuracy 97% and NB with lowest accuracy 90%. Fig.2. shows the ROC curve for different ML classification techniques applied on breast cancer dataset. Logistic regression classifier has highest AUC score 0.993 and Decision tree has lowest AUC score 0.938. Similarly Table 2 shows performance of ensemble techniques: RF, Adaboost, XGBoost on same dataset and Fig.3. shows the ROC curve for the same. XGBoost has achieved maximum accuracy 97% and highest AUC 0.999. Table 3 and Fig. 4. shows comparison with previous approaches used for breast cancer detection in terms of accuracy. Our ensemble model accuracy is slightly less than mentioned in [11] but Fig.5. shows the comparison of ROC curves between [11] and ours. Our XGBoost classifier achieved 97% accuracy and AUC-ROC 0.999.

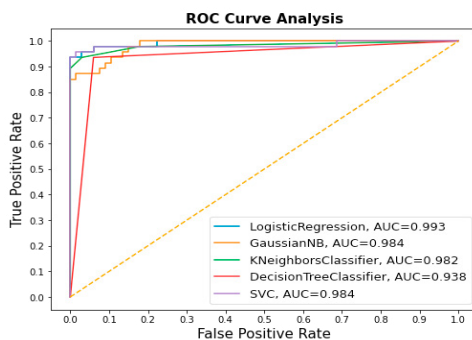


Fig.2. AUC-ROC Curve for ML Algorithm

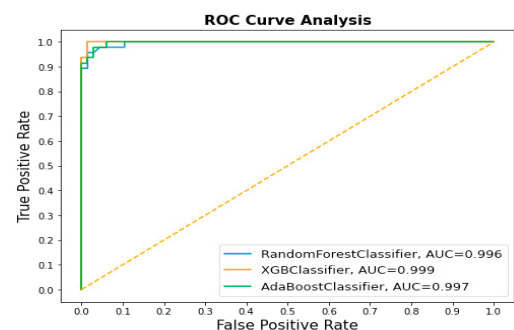


Fig.3. AUC-ROC Curve For Ensemble Techniques

Table 3: Comparison for Various Approaches on breast cancer prediction

Author	Technique	Accuracy	AUC
Naji et al.[11]	SVM	97.2	0.966
Amrane et al.[16]	MLP	95.4	-
Ara et al.[20]	RF	96.5	-
Chaurasia et al.[24]	J48	93.41	-
Ours	XGBoost	97	0.99

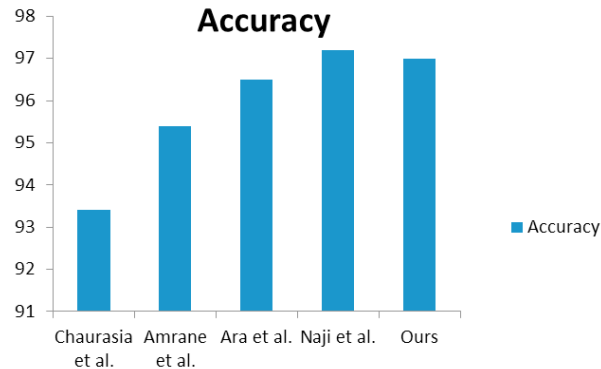


Fig.4. Accuracy comparison

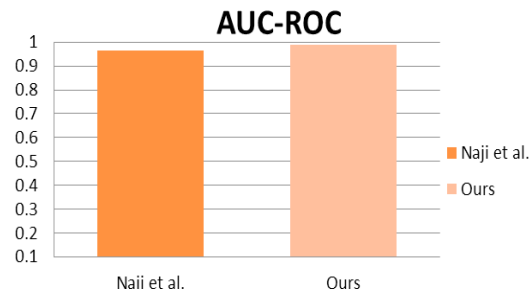


Fig.5. AUC-ROC Curve comparison with previous approach

5. Conclusion

Breast cancer is the important field of research and technology helps to reduce mortality rate caused by breast cancer. There are many ML algorithms introduced till now for analysis of medical datasets. It is essential for a medical diagnosis that the data on breast cancer be classified in a way that is both accurate and effective. Even though a many number of methods have been developed to classify breast cancer data, there are still many obstacles to overcome, including accuracy. In order to address this issue, we put forth a model for the classification of data relating to breast cancer. In this paper we applied six ML classification techniques: DT, KNN, SVM, RF, NB, and LR along with ensemble techniques on WDBC dataset. All techniques are compared and it has been found that in standard ML algorithm Decision Tree classifier with criterion gini index achieved highest accuracy: 97% and found highest AUC: 0.996 for LR classifier and for ensemble techniques XGBoost achieved highest accuracy 97% and AUC 0.99. It is possible that the proposed system for learning in ensembles would be of use to cancer specialists in the process of recognising cancer. The future scope of this work is to perform hyper parameter tuning to improve the performance of the model.

References

- [1] S Sung, H., Ferlay, J., Siegel, R. L., Laversanne, M., Soerjomataram, I., Jemal, A., & Bray, F. (2021). Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA: a cancer journal for clinicians*, 71(3), 209-249.
- [2] Ferlay, J., Colombet, M., Soerjomataram, I., Parkin, D. M., Piñeros, M., Znaor, A., & Bray, F. (2021). Cancer statistics for the year 2020: An overview. *International Journal of Cancer*, 149(4), 778-789.
- [3] Patil S, Kirange D, Nemade V. Predictive modelling of brain tumor detection using deep learning. *Journal of Critical Reviews*. 2020;7.
- [4] Nemade, V., Pathak, S., Dubey, A. K., & Barhate, D. (2022). A Review and Computational Analysis of Breast Cancer Using Different Machine Learning Techniques, 12(3)111-118.
- [5] Chaturvedi P, Jhamb A, Vanani M, Nemade V. Prediction and classification of lung cancer using machine learning techniques. In IOP Conference Series: Materials Science and Engineering 2021 Mar 1 (Vol. 1099, No. 1, p. 012059). IOP Publishing.
- [6] Abdelhafiz, D., Yang, C., Ammar, R., & Nabavi, S. (2019). Deep convolutional neural networks for mammography: advances, challenges and applications. *BMC bioinformatics*, 20(11), 1-20.
- [7] Nemade, V., Pathak, S., & Dubey, A. K. (2022). A Systematic Literature Review of Breast Cancer Diagnosis Using Machine Intelligence Techniques. *Archives of Computational Methods in Engineering*, 1-30.
- [8] Dora L, Agrawal S, Panda R, Abraham A.(2017) Optimal breast cancer classification using Gauss–Newton representation based algorithm. *Expert Systems with Applications*, 85:134-45
- [9] Obaid OI, Mohammed MA, Ghani MK, Mostafa A, Taha F. (2018) Evaluating the performance of machine learning techniques in the classification of Wisconsin Breast Cancer. *International Journal of Engineering & Technology*, 7(4.36):160-6.
- [10] Yellamma, P., Chowdary, C. S., Karunakar, G., Rao, B. S., & Ganesan, V. (2020). Breast Cancer Diagnosis Using MLP Back Propagation. *International Journal*, 8(9).
- [11] Naji, M. A., El Filali, S., Aarika, K., Benlahmar, E. H., Abdelouahid, R. A., & Debauche, O. (2021). Machine Learning Algorithms For Breast Cancer Prediction And Diagnosis. *Procedia Computer Science*, 191, 487-492.
- [12] Kabiraj, S., Raihan, M., Alvi, N., Afrin, M., Akter, L., Sohagi, S. A., & Podder, E. (2020, July). Breast cancer risk prediction using XGBoost and random forest algorithm. In *2020 11th international conference on computing, communication and networking technologies (ICCCNT)* (pp. 1-4). IEEE.
- [13] Liu, P., Fu, B., Yang, S. X., Deng, L., Zhong, X., & Zheng, H. (2020). Optimizing survival analysis of XGBoost for ties to predict disease progression of breast cancer. *IEEE Transactions on Biomedical Engineering*, 68(1), 148-160.
- [14] Nanglia, S., Ahmad, M., Khan, F. A., & Jhanjhi, N. Z. (2022). An enhanced Predictive heterogeneous ensemble model for breast cancer prediction. *Biomedical Signal Processing and Control*, 72, 103279.
- [15] Islam, M., Haque, M., Iqbal, H., Hasan, M., Hasan, M., & Kabir, M. N. (2020). Breast cancer prediction: a comparative study using machine learning techniques. *SN Computer Science*, 1(5), 1-14
- [16] Amrane, M., Oukid, S., Gagaoua, I., & Ensari, T. (2018, April). Breast cancer classification using machine learning. In *2018 electric electronics, computer science, biomedical engineerings' meeting (EBBT)* (pp. 1-4). IEEE.
- [17] Dhahri, H., Al Maghayreh, E., Mahmood, A., Elkilani, W., & Faisal Nagi, M. (2019). Automated breast cancer diagnosis based on machine learning algorithms. *Journal of healthcare engineering*.
- [18] Kharya S, Soni S. Weighted naive bayes classifier: a predictive model for breast cancer detection. *International Journal of Computer Applications*. 2016 Jan;133(9):32-7
- [19] Huang Q, Chen Y, Liu L, Tao D, Li X. On combining biclustering mining and AdaBoost for breast tumor classification. *IEEE Transactions on Knowledge and Data Engineering*. 2019 Jan 14;32(4):728-38.
- [20] Ara S, Das A, Dey A. Malignant and benign breast cancer classification using machine learning algorithms. In 2021 International Conference on Artificial Intelligence (ICAI) 2021 Apr 5 (pp. 97-101). IEEE.
- [21] Jabbar MA. Breast cancer data classification using ensemble machine learning. *Engineering and Applied Science Research*. 2021 Jan 27;48(1):65-72.
- [22] Mahesh TR, Vinoth Kumar V, Vivek V, Karthick Raghunath KM, Sindhu Madhuri G. Early predictive model for breast cancer classification using blended ensemble learning. *International Journal of System Assurance Engineering and Management*. 2022 Jun 26:1-0.
- [23] [https://archive.ics.uci.edu/ml/datasets/breast+cancer+wisconsin+\(diagnostic\)](https://archive.ics.uci.edu/ml/datasets/breast+cancer+wisconsin+(diagnostic))
- [24] Chaurasia V, Pal S, Tiwari BB. Prediction of benign and malignant breast cancer using data mining techniques. *Journal of Algorithms & Computational Technology*. 2018 Jun;12(2):119-26.